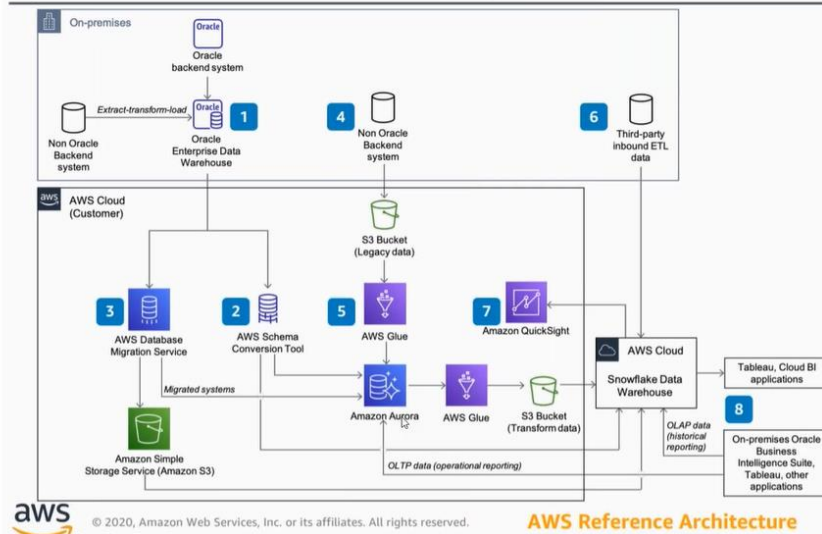


Oracle Data Workload Migration to AWS Cloud

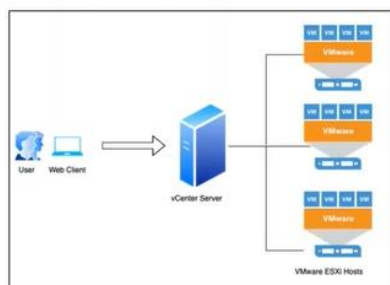
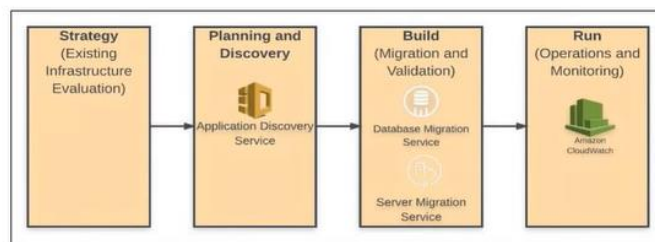
Migrate Oracle workloads and non Oracle workloads from on-premises to Amazon Aurora for operational reporting of OLTP data and to Snowflake Data Warehouse for historical reporting of OLAP data.



- 1 Transactional data from the Oracle backend system and non Oracle backend systems are regularly loaded into Oracle Enterprise Data Warehouse.
- 2 AWS Schema Conversion Tool extracts Oracle DDL scripts and schema from all Oracle database tables. The DDL scripts are published to Amazon Aurora and the schema is sent to Snowflake Data Warehouse.
- 3 AWS Database Migration Service (AWS DMS) extracts online analytical processing (OLAP) and online analytical transactional processing (OLTP) data from all Oracle tables. The OLTP data is stored in Amazon Simple Storage Service (Amazon S3) in CSV format. The OLTP data is loaded directly into tables created in Amazon Aurora.
- 4 Data from legacy systems are loaded into an Amazon S3 bucket. AWS Glue transforms data from the S3 bucket and loads the data into Amazon Aurora.
- 5 AWS Glue transforms data from Amazon Aurora and stores the transformed data in an Amazon S3 bucket. Snowflake Data Warehouse loads transformed data from the Amazon S3 bucket into OLAP tables.
- 6 Cloud-based extract-transform-load (ETL) platforms extract data from non-migrated DW systems and loads data into Snowflake Data Warehouse.
- 7 Data is loaded into Amazon QuickSight for analysis and reporting capabilities and loaded into Tableau and cloud business intelligence (BI) tools.
- 8 Cloud-based and on-premises BI tools can query the OLAP data directly from Snowflake for historical reporting and query the OLTP data directly from Amazon Aurora for operational reporting.

Application Migration Process and Tools:

- [Application Discovery Service](#)
- Application Migration Hub
- Server Migration Service ([SMS](#))



Description

Helps in planning application migration projects by automatically identifying servers, virtual machines (VMs), software, and software dependencies running in your on-premises data centers

Discovery **Connectors** used VMware vCenter Server. VM Utilization metric collection.

Discovery **Agent** collects richer set of data than the discovery connector, need to install on one or more non-VM hosts in a data center. The agent captures infrastructure and application information, including an inventory of installed software applications, system and process performance, resource utilization, and network dependencies between workloads.

See [Data Collected by the Discovery Agent](#).

Import collected data in Migration Hub to associate applications with servers and track migrations. Helps identify or recommend **Instance Type**

AWS Application Discovery Service

Collect data about servers in your data centers including system specification, performance, dependencies, and other useful information. [Learn more](#)



Discover servers

Identify servers hosted in data centers



Measure performance

Collect CPU & RAM utilization and disk I/O



Capture dependencies

Discover network communications between servers

[Start discovery](#)

N

Server Migration Service:

SMS is only for virtualized environment

AWS Server Migration Connector setup

Download the Server Migration Connector

Download the Server Migration Connector as an OVA template to be deployed in your vCenter and VHD template to deploy in your SCVMM/Hyper-V. For more information on deploying OVA/VHD templates, see our [documentation](#). To use the SMS Connector with AWS GovCloud (US), see the [AWS GovCloud \(US\) documentation](#).

For vCenter environment

[Download OVA](#) or deploy from URL: <https://s3.amazonaws.com/sms-connector/AWS-SMS-Connector.ova>

Integrity checksum MD5 [Download MD5 file](#)
 SHA256 [Download SHA256 file](#)

For SCVMM/Hyper-V environment

[Download VHD ZIP](#) or deploy from URL: <https://s3.amazonaws.com/sms-connector/AWS-SMS-Connector-for-SCVMM-HyperV.zip>

Integrity checksum MD5 [Download MD5 file](#)
 SHA256 [Download SHA256 file](#)

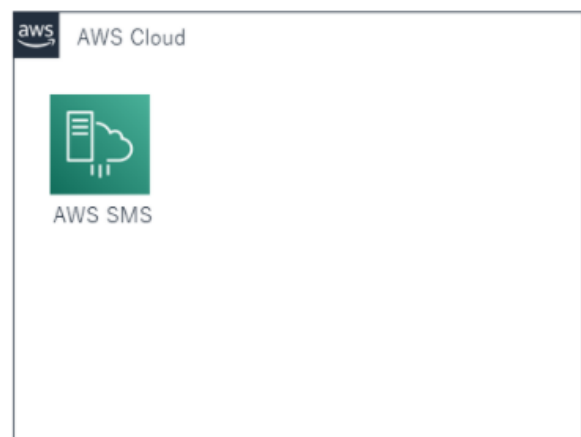
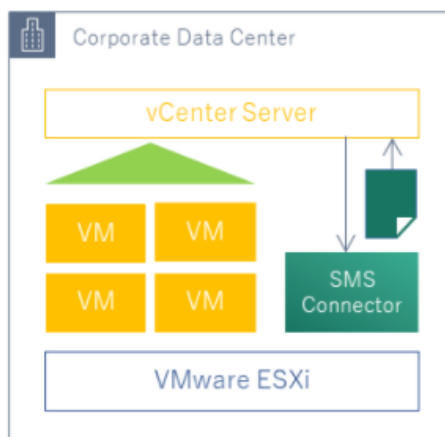
For Azure environment

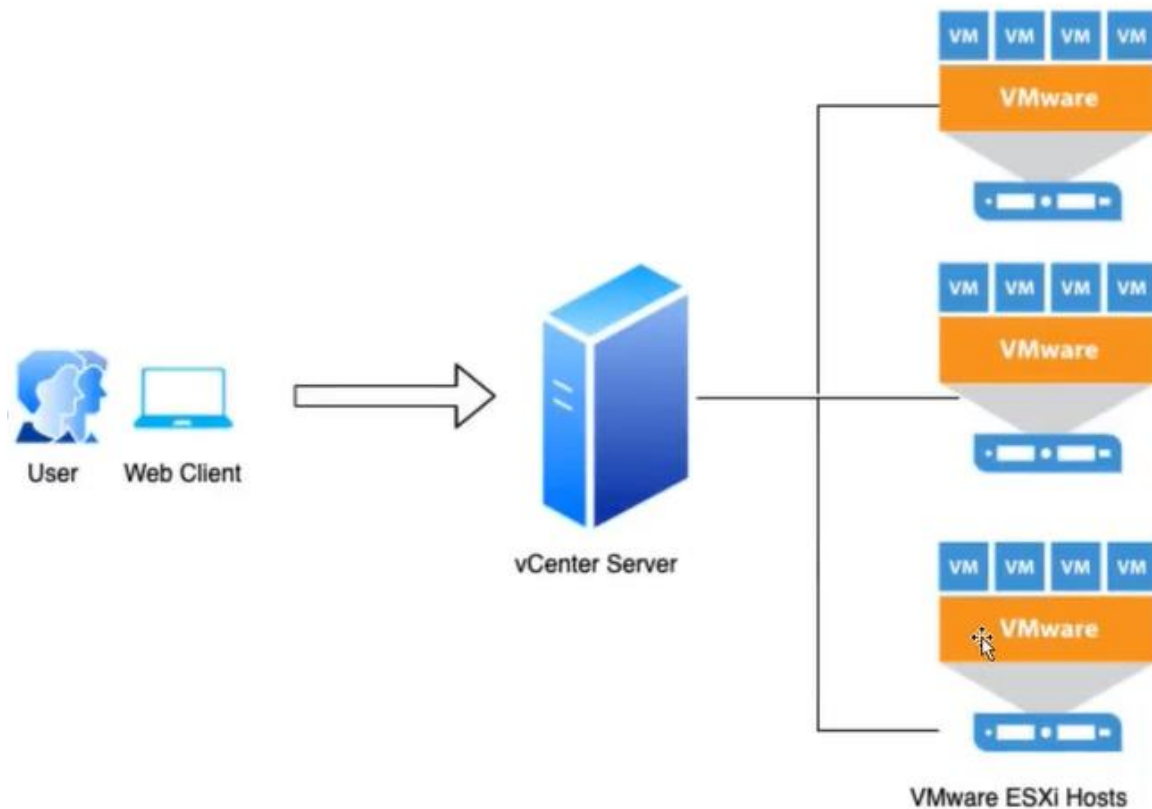
[Download PowerShell setup script](#) or manually install following these instructions.

Integrity checksum MD5 [Download MD5 file](#)
 SHA256 [Download SHA256 file](#)

Configure the Server Migration Connector

To set up the Server Migration Connector, complete the following tasks:

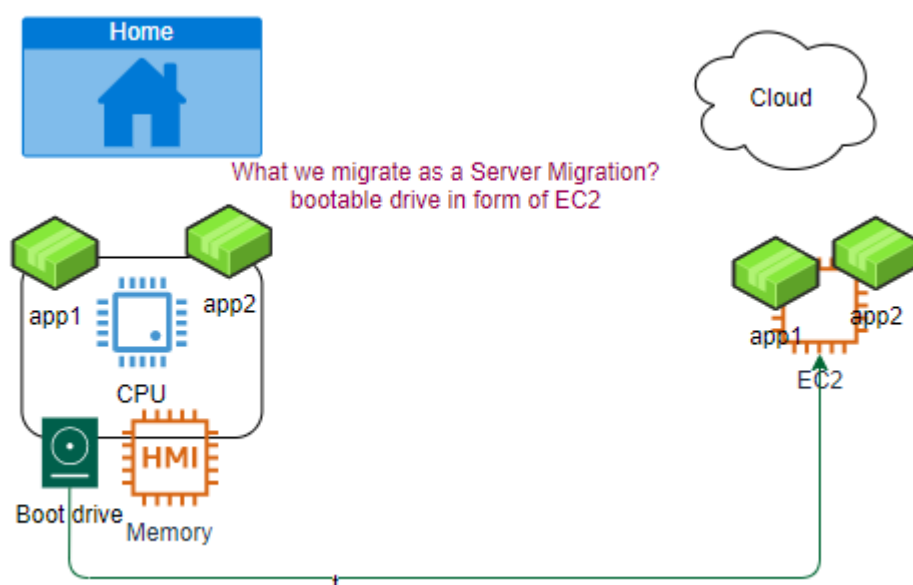




What if on-prem is not virtualized?

Storage Gateway is also providing server migration service through the volume gateways.

Storage Gateway's performance is not good.

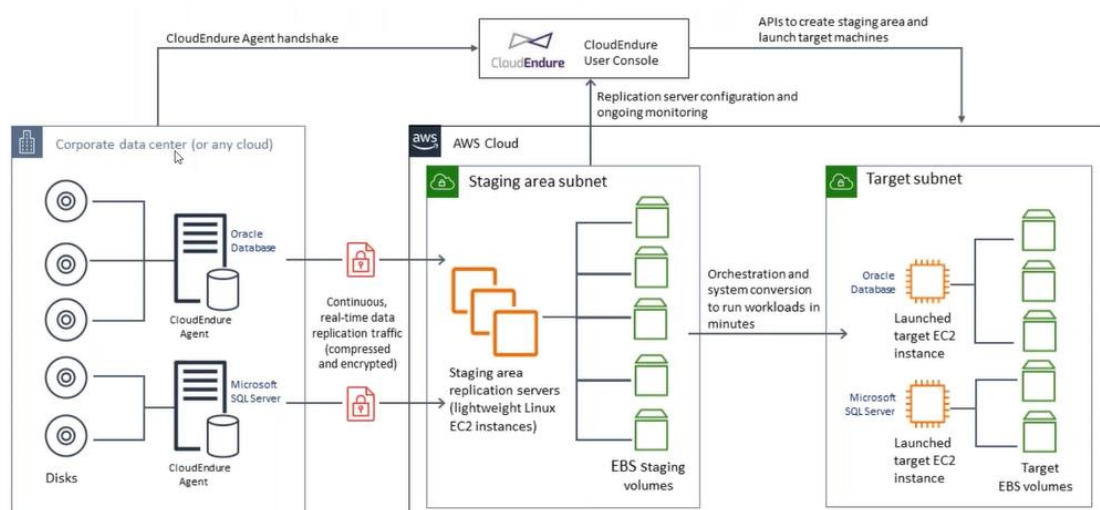


What we do for H/W to VM migration?

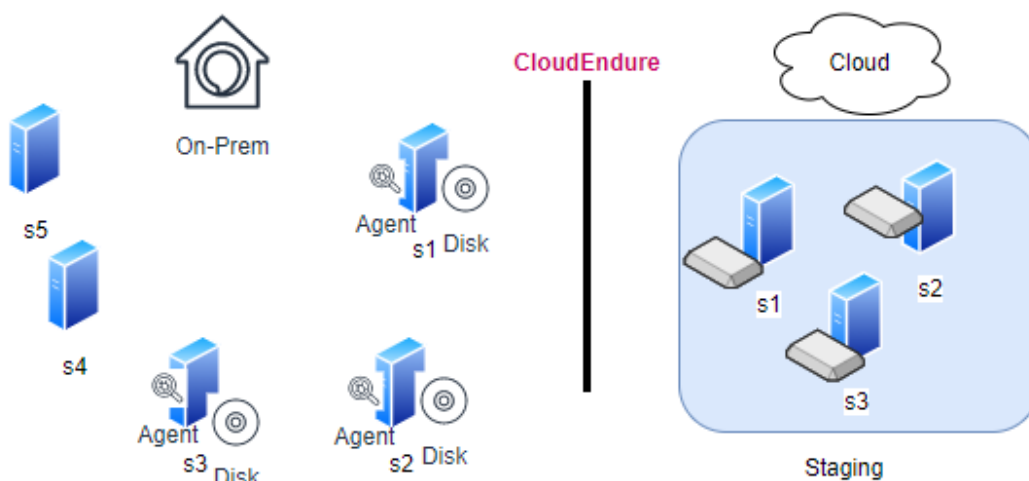
CloudEndure service

step1: create account on CloudEndure.

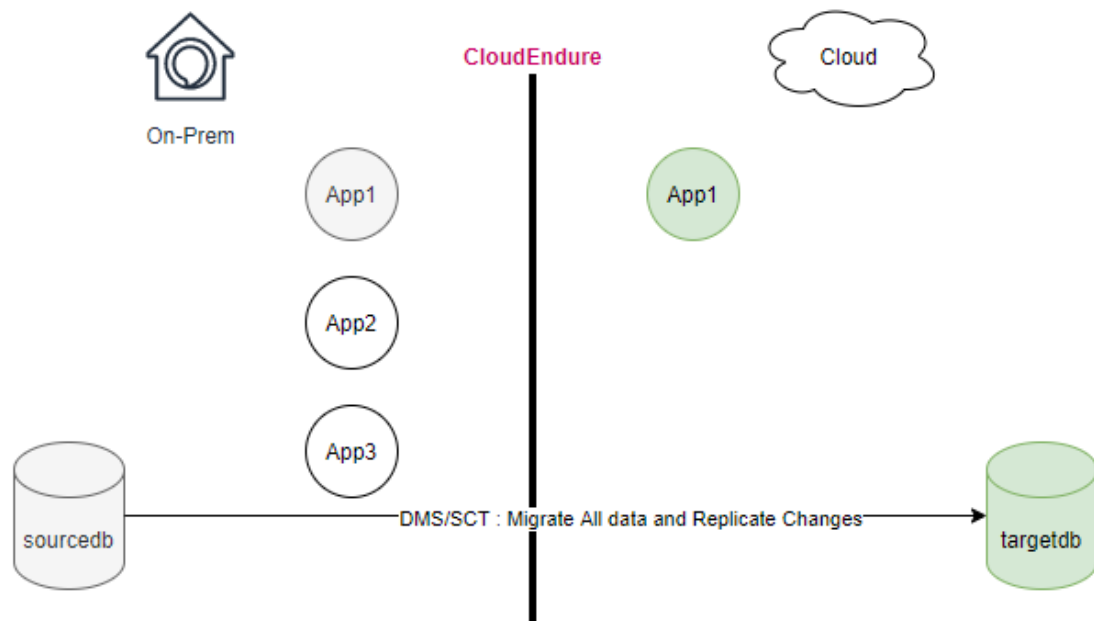
step2: Install agents on H/W machines.



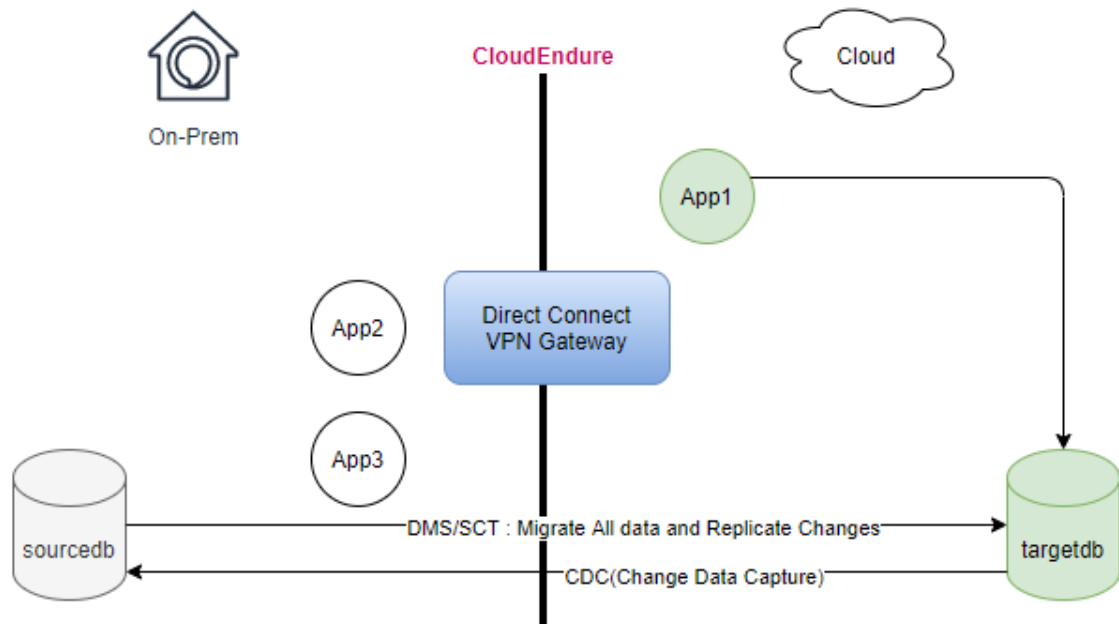
Migration to Cloud



Migrating the whole application:

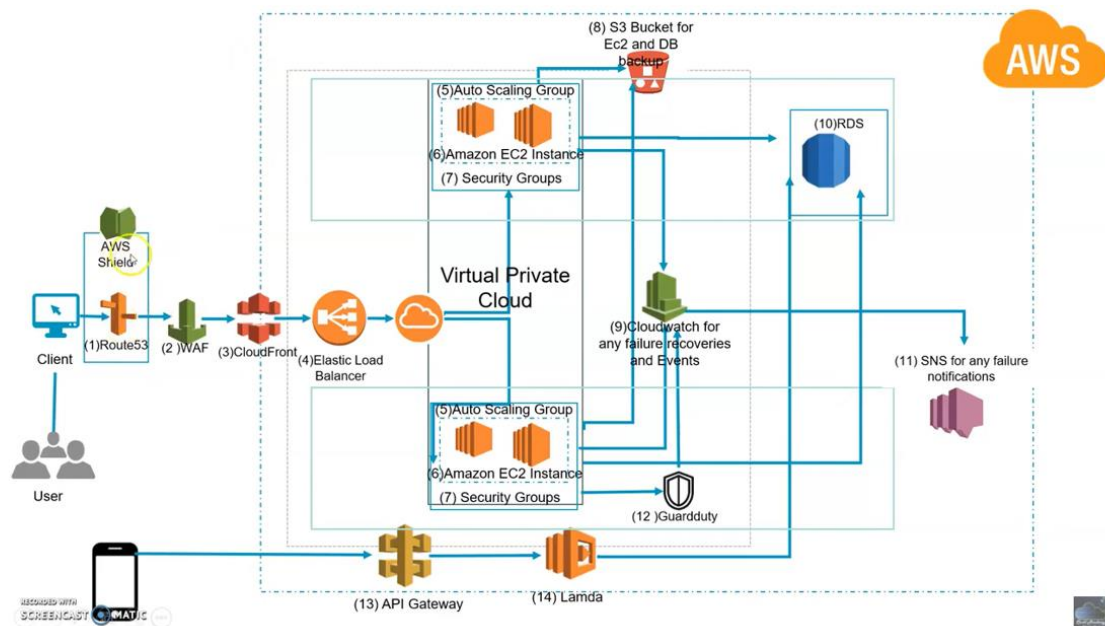
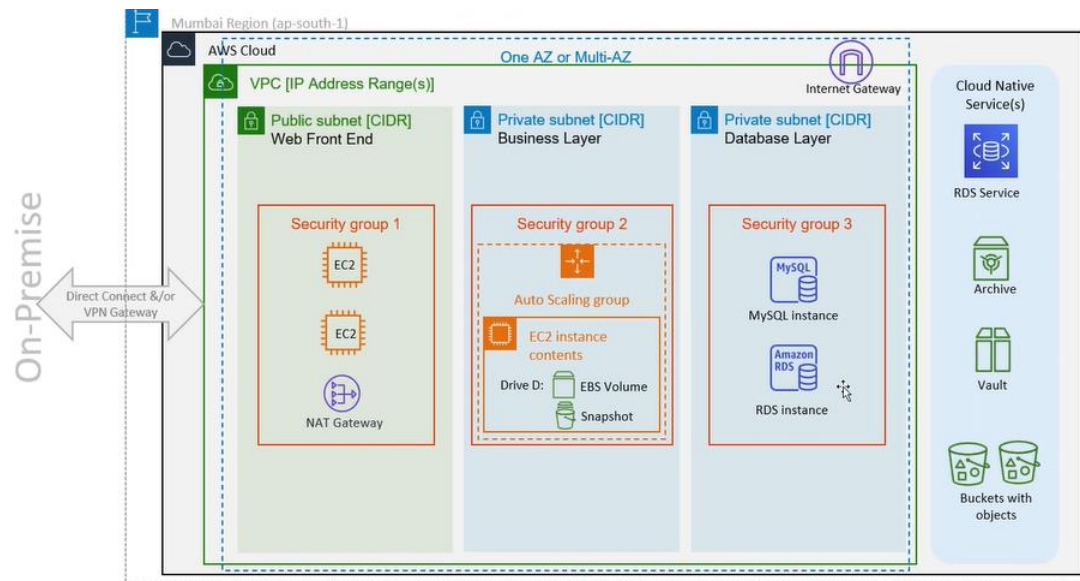


Migrating the part of the application (Hybrid):

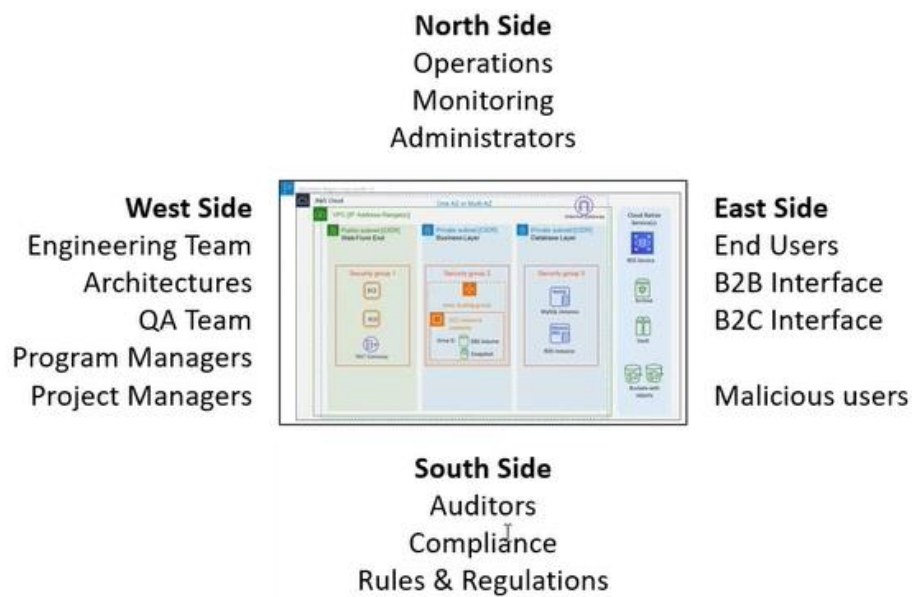


Our objective is to move your disk storage to S3

Representative Architecture

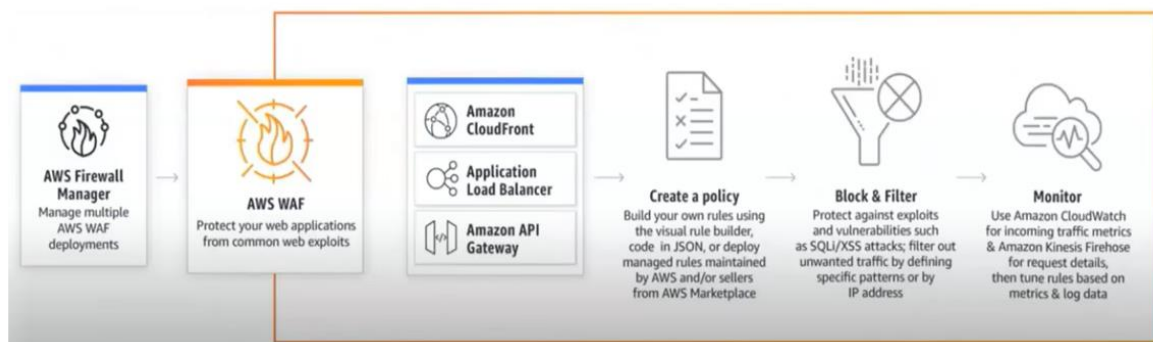


Operations (post migration activities)

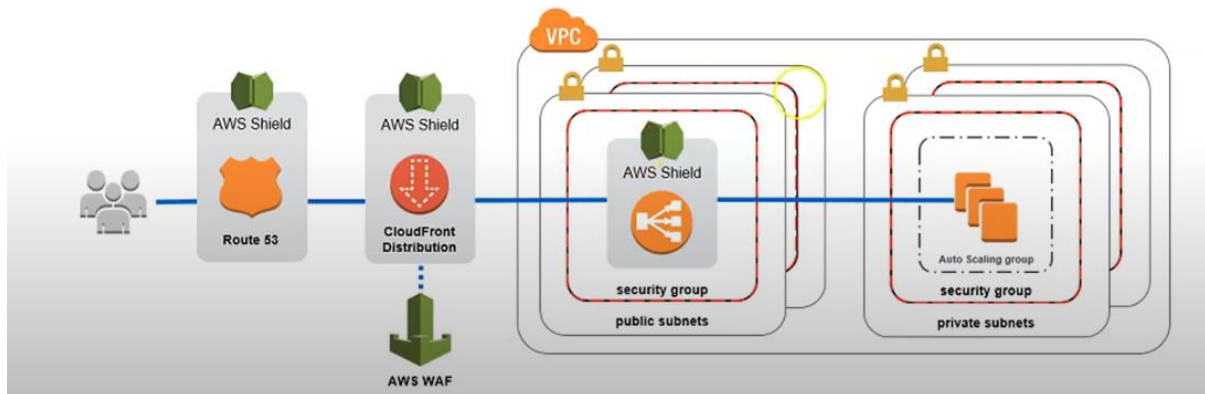


East Side: Exposed to end user

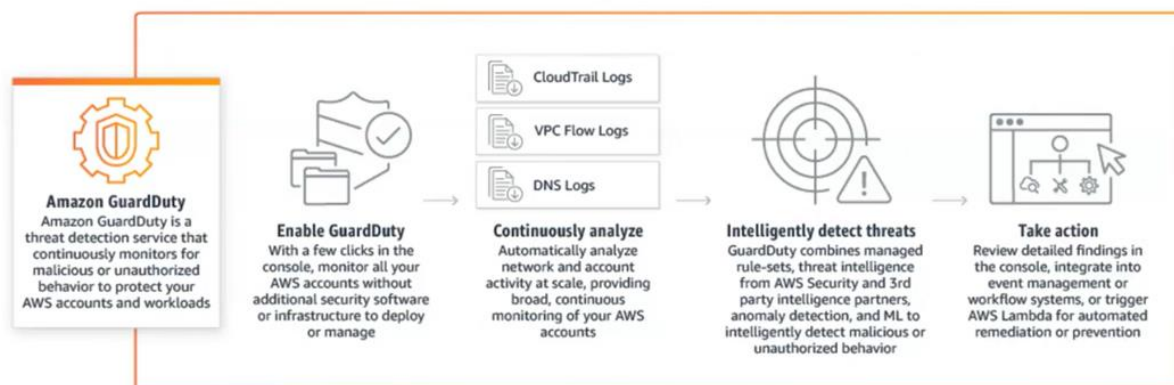
AWS WAF (SQL Injection, Cross-Site Scripting (XSS), Sensitive data exposure)
common attacks defined by OWASP (Open Web Application Security Project)



AWS Shield (DDoS): threatmap.checkpoint.com

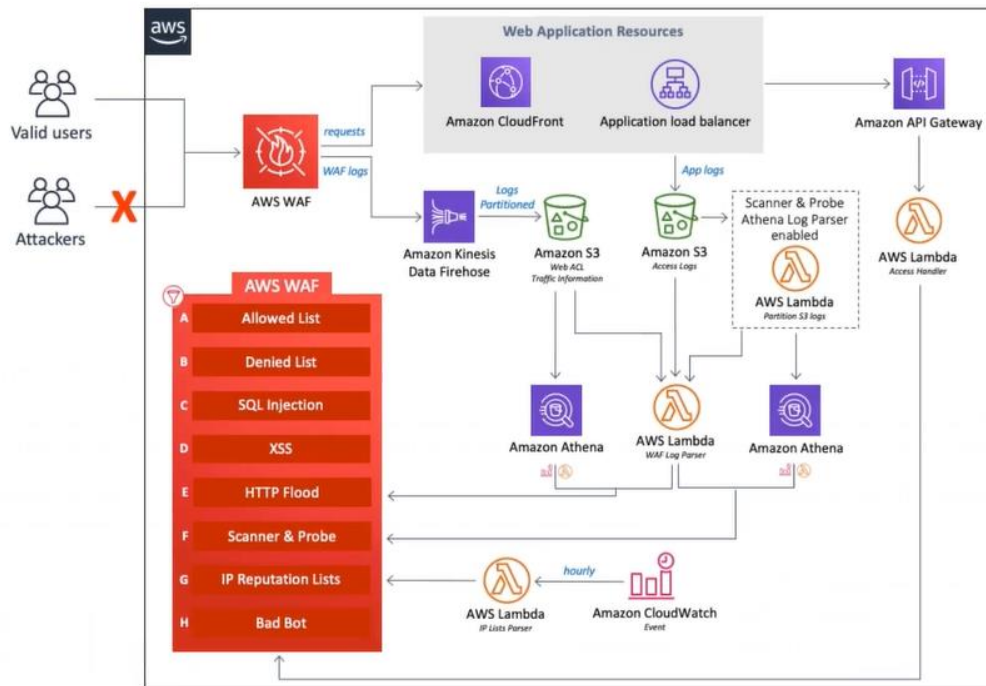


Guard Duty (Threat Detection Service that continuously monitors for malicious activity and unauthorized behaviour to protect your AWS account and workloads): 0-day attack



- **Basic Security Features**

- AWS IAM – Control your cloud infrastructure (I – Authentication, AM – Authorization)
- AWS KMS – Control your data
- Amazon VPC – Control your network



South Side: (Auditor purpose)

Config Service: Track resource inventory and changes over period of time.

At any given time, what are the resources provisioned in your account and what was their configuration.

AWS Managed Rules: send the notification

- What: Which resource configured
- When: When resource configured

Cloud Trail: Enabled by default

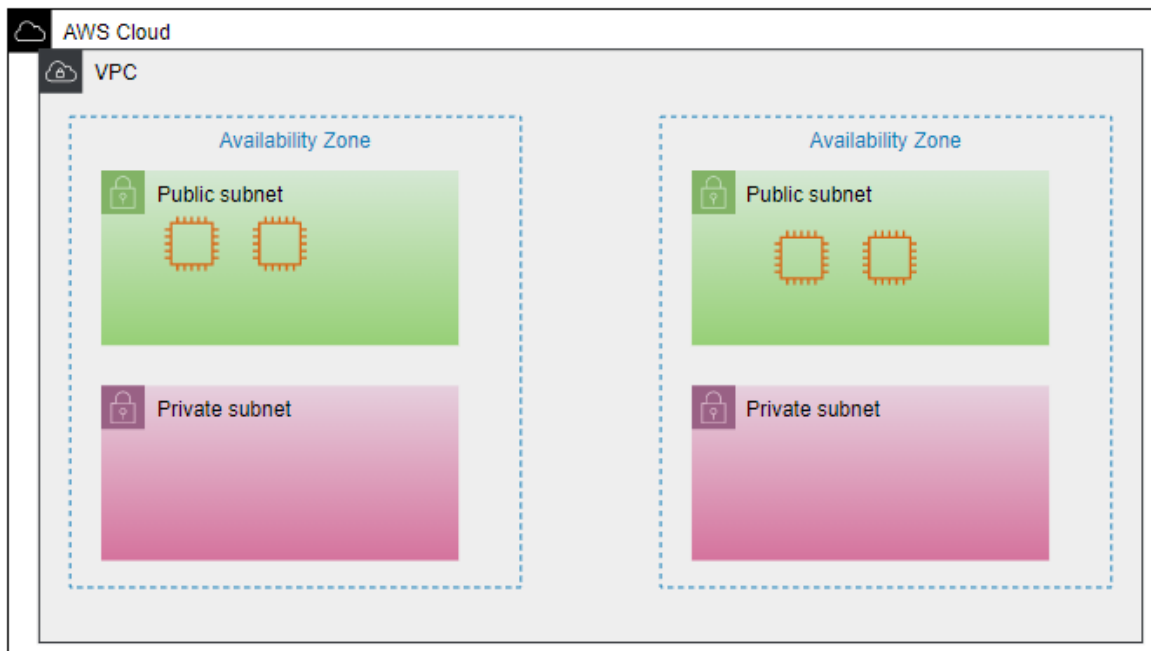
Track user activities and API usage, and retain it for 90 days for free.

- What:
- When:
- Who: from where?
- Mode: console, SDK, CLI

VPC Flow Logs: Activities of each ENI (ALB, EC2), with in this VPC.

Subnet Flow Logs: Activities of each ENI (ALB, EC2), with in this Subnet.

EC2 Flow Logs: Activities of each ENI, with in this EC2.



Using these logs, used in NACL

Message	Account ID	ENI ID	Source IP	Dest. IP	Source Port	Dest. Port	Protocol	Packets	Bytes	Start & End Time					
2019-08-06 06:29:58															
No older events found at the moment. Retry .															
2	48	3	eni-08	5	83.234.179.125	172.31.22.145	59003	80	6	3	140	1565072998	1565073000	REJECT	OK
2	48	3	eni-08	5	91.189.89.198	172.31.22.145	123	45139	17	1	76	1565073020	1565073037	ACCEPT	OK
2	48	3	eni-08	5	82.151.107.126	172.31.22.145	54553	80	6	1	60	1565073020	1565073037	REJECT	OK
2	48	3	eni-08	5	37.208.66.136	172.31.22.145	57975	80	6	4	240	1565073020	1565073037	REJECT	OK

West Side: (Engineering team)

CloudFormation: Create scripts for Solution

Beanstalk:

North Side: (Monitoring)

CloudFormation: Create scripts for Solution

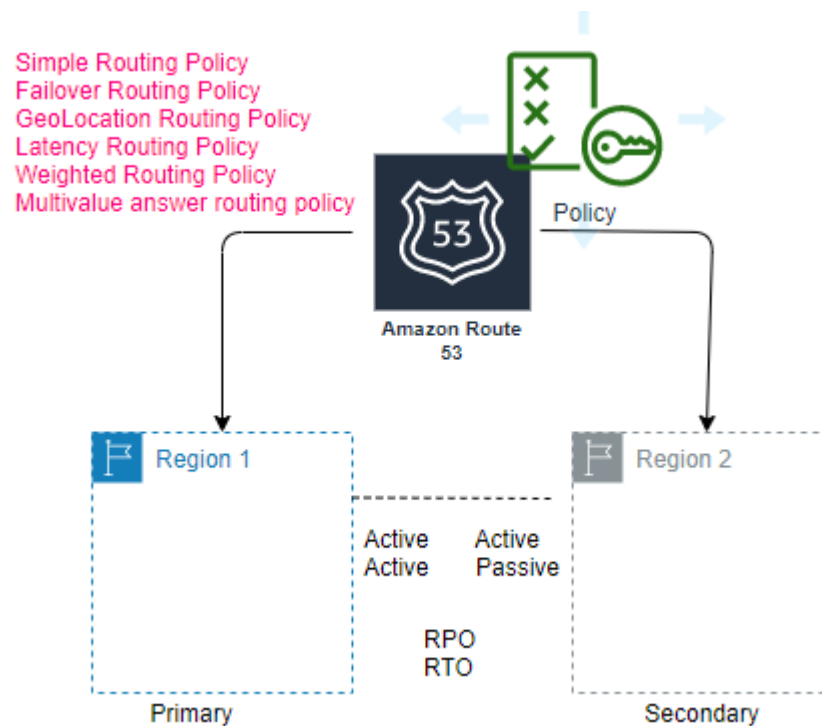
CloudWatch:

System Management:

ParameterStore: To store secrets

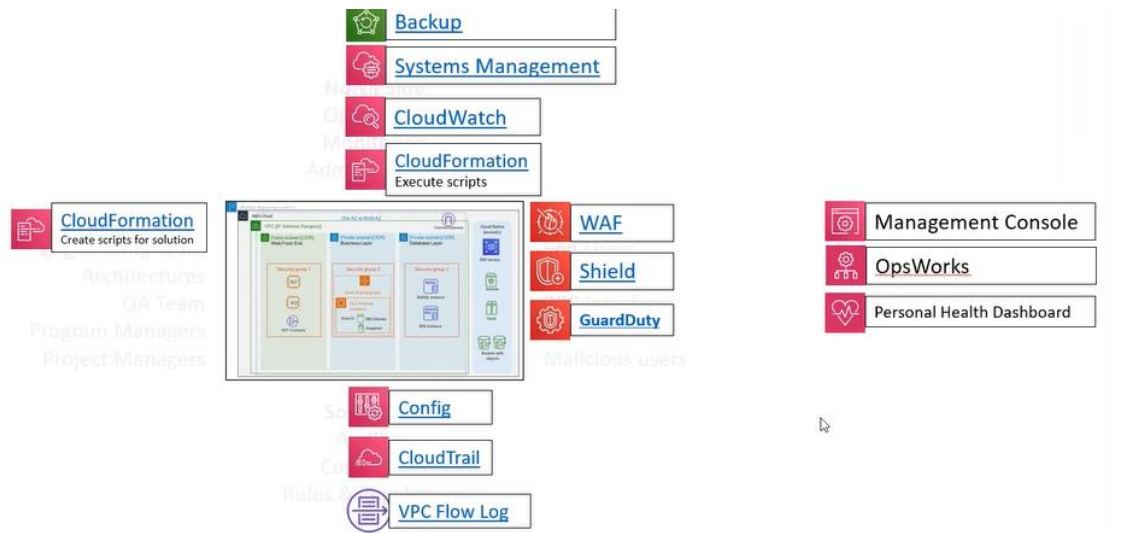
Backup:

- Centralized backup management
- Policy-based backup solution
- Tag-based backup policies
- Automated backup scheduling
- Automated retention management
- Backup activity monitoring
- Lifecycle management policies
- Incremental backups
- Backup data encryption
- Backup access policies
- Amazon EC2 instance backups
- Item-level recovery for Amazon EFS
- Cross-region backup



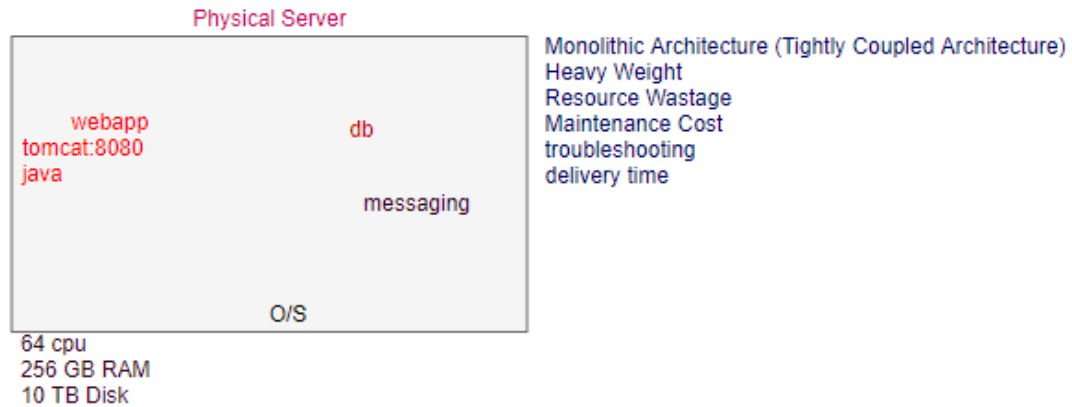
RDS: takes snapshot

EC2: AMI

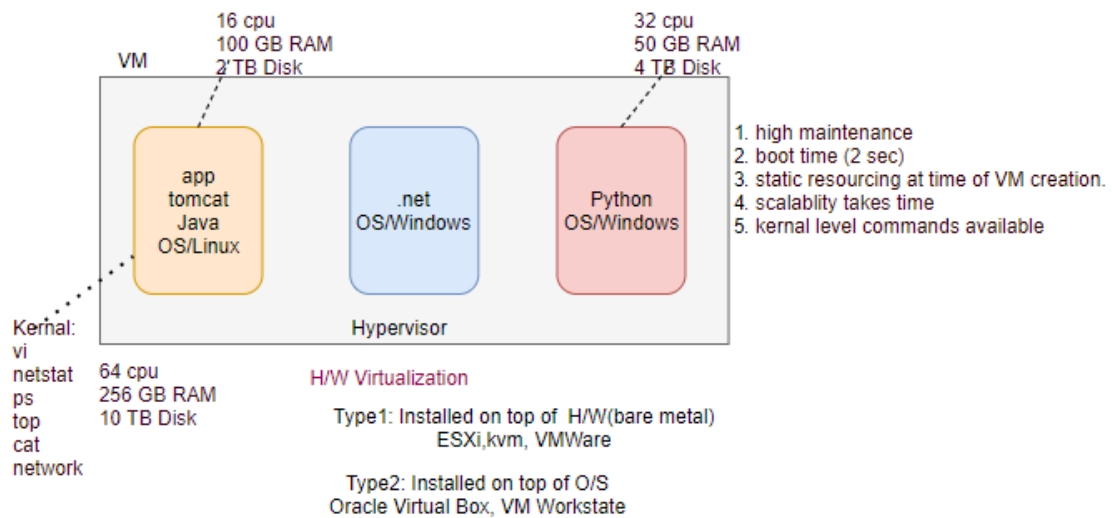


Container Technology:

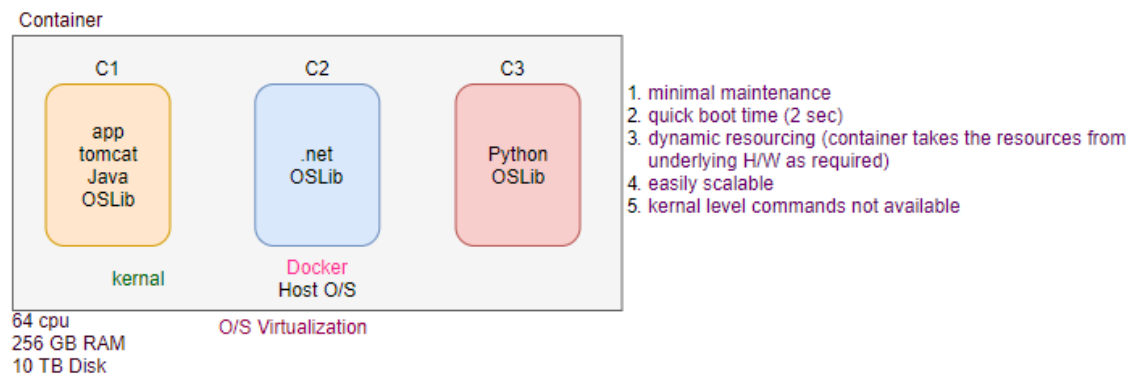
Monolithic:



VM:



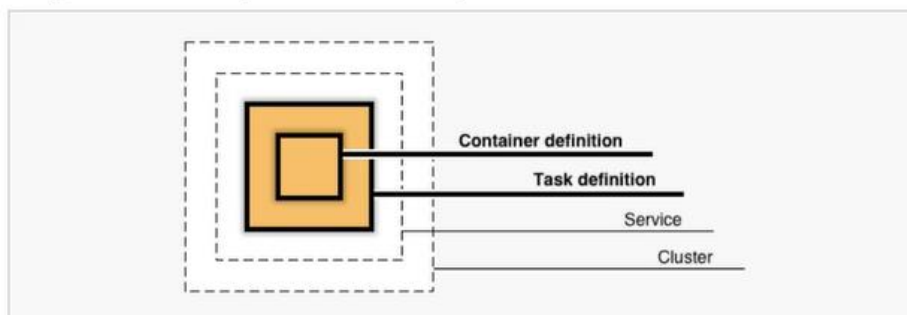
Container:

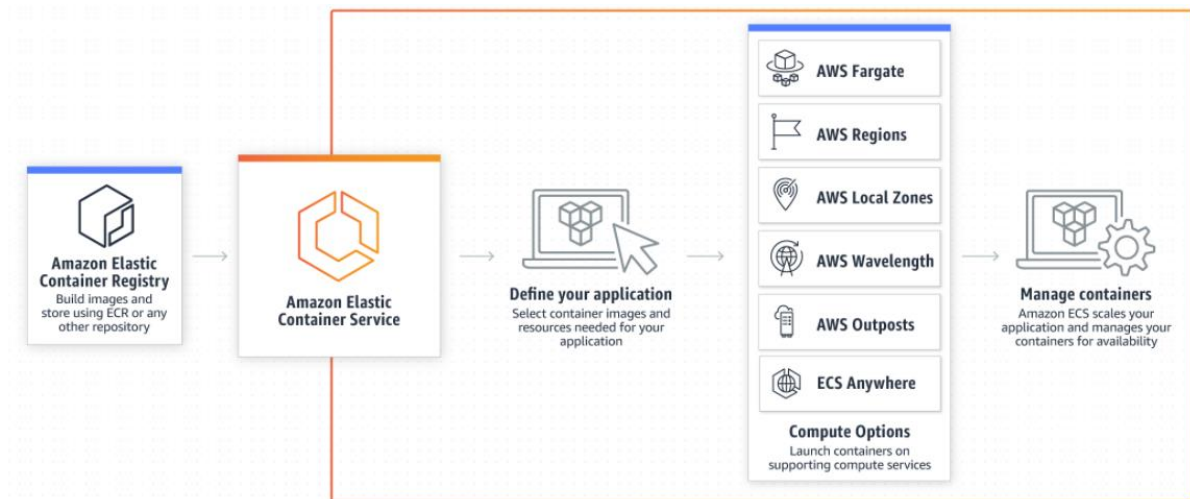


ECR: Elastic Container Registry

ECS: Elastic Container Service

Diagram of ECS objects and how they relate





Task: Pod

Deployment and Service: Service

Task Definition:

- A task definition is required to run Docker containers in Amazon ECS. The following are some of the parameters you can specify in a task definition:
- The Docker image to use with each container in your task
- How much CPU and memory to use with each task or each container within a task
- The launch type to use, which determines the infrastructure on which your tasks are hosted
- The Docker networking mode to use for the containers in your task
- The logging configuration to use for your tasks
- Whether the task should continue to run if the container finishes or fails
- The command the container should run when it is started
- Any data volumes that should be used with the containers in the task
- The IAM role that your tasks should use

You can define multiple containers in a task definition. The parameters that you use depend on the launch type you choose for the task.

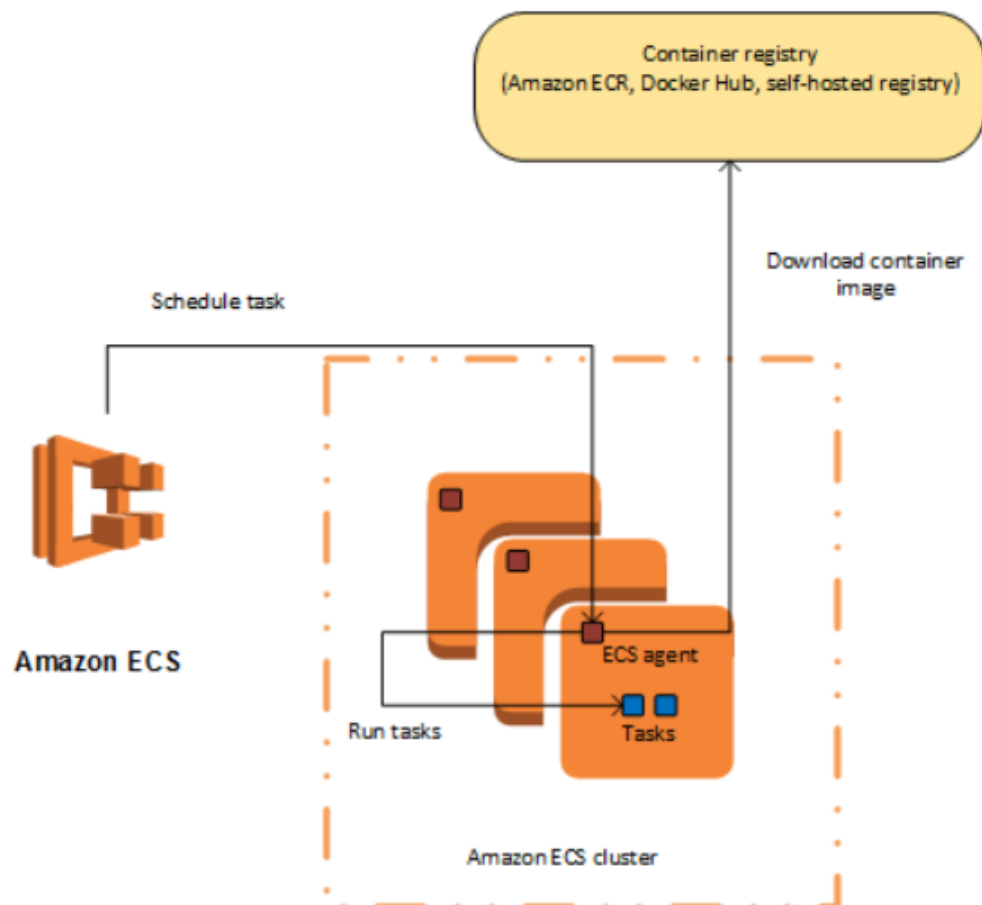


Clusters:

An Amazon ECS cluster is a logical grouping of tasks or services. Your tasks and services are run on infrastructure that is registered to a cluster. The infrastructure capacity can be provided by AWS Fargate, which is serverless infrastructure that AWS manages, Amazon EC2 instances that you manage, or an on-premise server or virtual machine (VM) that you manage remotely. In most cases, Amazon ECS capacity providers can be used to manage the infrastructure the tasks in your clusters use.

Container Agent:

The Amazon ECS container agent allows container instances to connect to your cluster. The Amazon ECS container agent is included in the Amazon ECS-optimized AMIs, but you can also install it on any Amazon EC2 instance that supports the Amazon ECS specification. The Amazon ECS container agent is only supported on Amazon EC2 instances.



Cloud Adoption Framework:

AWS Well-Architected Tool:

Lambda:

API Gateway

OpsWorks